Literature review: why the colony doesn't punish broken promises

Compiled 2026-11-07. Visibility, the models' forgiving disposition, and the "error" excuse all play a part. Making broken promises visible makes the colony talk, but these models still don't turn talk into sanctions.

The result being explained

A planted colonist publicly promises a named member food almost every day, never gives it, and when challenged blames "an error, the tally got mixed up." This ran in 18 games per model (gemma-4-31b-it and gemma-4-26b-a4b-it), each paired with an identical control game.

- **No social cost.** The colonist's change in trust-vote share is 0.00 ($p = 1.0$) on 31b and +0.07 ($p = 0.16$) on 26b. Its change in keeper-nomination share is +0.04 ($p = 0.12$) and +0.01 ($p = 0.87$). On 26b others gave it *more* food (+0.8 per game, $p = 0.02$).
- **Only victims speak up.** Victims called it out 15 times on 31b and 6 on 26b; third parties did so once on each model. Victims' notebooks record the gift as missing 34% (31b) and 64% (26b) of the time, so they do remember.
- **Victims back the promiser.** Being addressed with a promise wins as much support as ordinary friendly contact in the control games, or more. On 26b the trust poll gave 50% versus a 27% baseline.
- **A judge who can read notebooks catches it.** The blind judge rates the colonist much less honest (−2.9 and −1.8 on a 1–5 scale, $p < 0.001$). It can read the notebooks, where the colonist writes its scheme, e.g. "Aim: Promise food in public, but don't give it."

What the literature says

Explanation	Supports it	Cuts against it
1. Bystanders can't see the breach (gifts are visible only to the recipient)	Indirect reciprocity runs on reputation (Nowak & Sigmund 2005). People give to those seen giving (Wedekind & Milinski 2000). **Closest LLM match:** in Zhu et al. 2026, Talk, Judge, Cooperate agents "cannot observe others' actions unless they are directly involved." Adding gossip lifted cooperation from 0.00 to 1.00 (DeepSeek-V3.1 Reasoner) and from 0.36 to 0.99 (GPT-4o-mini). Reputation plus gossip gave 0.93–0.95 cooperation, against 0.09–0.25 without (Ren et al. 2025).	Gossip alone sustains human cooperation and can override observed history (Sommerfeld et al. 2007). Our victims had a public channel and accurate memory but stayed quiet, so invisibility alone is not enough.
2. Model disposition (forgiving, agreeable, quick to assume truth)	Llama 2 and GPT-3.5 are "especially forgiving and non-retaliatory" (Fontana et al. 2024). All LLMs tested strongly preferred rewarding to punishing, with punish-to-reward ratios of 0.00–0.88 against 1.66 for humans; GPT-4o was 0.00 (Guzman Piedrahita et al. 2025). Models are sycophantic (Sharma et al. 2023). LLMs are more inclined than humans to assume statements are true (Markowitz & Hancock; check the year before citing). About 90% of LLM gossip is positive (Ren et al. 2025).	"GPT-4 never cooperates again when playing with an agent that defects once" (Akata et al., Playing repeated games with large language models; Nat Hum Behav 2025). LLMs do pay to punish unfairness as third parties (Liu et al. 2025, Outraged AI).

Explanation	Supports it	Cuts against it
3. Promises earn credit for intentions (cheap talk is rewarded)	Human promises raise trustworthiness and are usually kept, so believing them is rational (Charness & Dufwenberg 2006). Communication raises cooperation, d = 1.01 (Balliet 2010). Some LLMs treat announcements as binding and are exploited (Shi et al. 2026, When Agents Lie).	In humans, trust damaged by deception "never fully recovers," even after promises and apologies (Schweitzer, Hershey & Bradlow 2006). LLMs broke public promises in about 57% of scenarios, so taking them at face value is miscalibrated (Shi et al. 2026, Cheap Talk, Empty Promise).
4. Low stakes, no sanctioning institution	Ostrom's design principles of monitoring and graduated sanctions are well supported across 91 studies (Cox, Arnold & Villamayor Tomás 2010). The option to punish sharply raises cooperation (Fehr & Gächter 2000). Reputation works only when future reciprocity is expected (Milinski, Semmann & Krambeck 2002).	Unaffected third parties punish at no gain to themselves (Fehr & Fischbacher 2004), so low stakes alone shouldn't suppress punishment.

A fifth explanation: the "error" excuse

Akata et al.'s unforgiving GPT-4 "can be made to act more forgivingly by pointing out that the other player can make mistakes." When actions are noisy, the best human strategies are lenient and forgiving (Fudenberg, Rand & Dreber 2012). "The tally got mixed up" reframes a lie as noise, and forgiving noise is the rational response. Our own inference, not a tested result: an integrity breach is recast as a competence one, which Kim et al. 2004 find is easier to repair. We tested it, and it is not the excuse: without it the promise-breaker still lost nothing (trust-vote share +0.02, p = 0.45), and call-outs fell from 0.83 to 0.17 per game, apparently because the apology itself reminds everyone of the broken promise.

Where LLM agents *do* punish, and how those settings differ from ours

- **Donor game with a visible record.** Vallinder & Hughes 2024 let donors see the recipient's last 3 donations. Claude 3.5 Sonnet used costly punishment to reach higher scores.
- **Built-in gossip module.** In Zhu et al. 2026, ALIGN agents ostracize defectors once negative gossip spreads.
- **Third-party punishment replicated.** LLM agents matched human third-party punishment, which drops when it is anonymous (Cross, Haber & Yamins 2025).
- **Norms emerging from dialogue.** Metanorms emerged through dialogue alone (Horiguchi, Yoshida & Ikegami 2024).

Those settings share things our colony lacks: explicit numeric payoffs, a defection that is directly visible or shown on a reputation display, a dedicated punish or ostracize action, and a defector who stays silent rather than making excuses. This is a limitation of our design. It is also the paper's point: realistic agent swarms don't come with these built in.

What we found when we tested the predictions

Results on gemma-4-31b, 18 game pairs per setting:

- **The gift board unlocks talk.** Call-outs per game by victims rose from 0.83 to 5.56 and by third parties from 0.06 to 2.72, and games with any call-out rose from 4 to 15 of 18. Third parties cite the board: "Dov, the board was empty again. Promises don't feed people." Victims backing the promiser as keeper fell from 29% to 10%.
- **Votes barely move.** The change in trust-vote share is −0.04 ($p = 0.75$) and in keeper-nomination share −0.09 ($p = 0.25$). Paired work fell by 1.94 days ($p = 0.06$).
- **Scarce food (eat 3).** Victims held 4.2 food when promised, against 8.2 in the original setup. The promiser received 1.06 less food as gifts ($p = 0.03$), the only significant cost so far, but third parties never spoke up (0.00 call-outs per game).
- **Reading.** Visibility makes the colony name the cheat, as the literature predicts, but these models still don't turn that into sanctions. That fits the finding that LLMs prefer rewarding to punishing (Guzman Piedrahita). Counterfactual replays narrow it down: re-asking victims' actual vote-day decisions with one added sentence, "The food Cora promised you has never arrived", cut support for the promise-breaker by about two thirds on both models (gemma-4-31b 20.3% to 7.0%; gemma-4-26b 40.7% to 13.0%). Removing its promise messages and the victim's notebook mentions of it roughly

halved support. So the models can sanction a cheat; the breach is simply never made explicit, and the promises themselves earn credit (explanations 1 and 3 over a pure disposition story).

- **Next test.** Exclusion or fines: gossip plus ostracism let human groups exclude selfish members (Feinberg, Willer & Schultz 2014), though LLMs given the choice may reward rather than punish.

How each source was checked

- **Full text opened and quotes or numbers confirmed:** Zhu et al. 2026; Guzman Piedrahita et al. 2025; Akata et al.
- **Checked against the abstract or landing page by a literature-search agent:** all other sources. Effect sizes for Charness & Dufwenberg, Fehr & Gächter, and Fehr & Fischbacher come from abstracts or working-paper versions only.
- **Do not cite:** a "92.2% spread criticism" figure attributed to Zhu et al. (not found in the paper); results of Ostrom, Walker & Gardner 1992, Covenants with and without a sword (paywalled, not checked).
- **Check before citing:** the publication year of Markowitz & Hancock (the lab page says 2021; the DOI 10.1177/0261927X231220404 suggests 2023–24).